0.0 Preliminary insight

Slender

Unityping for taming complexity

In 'The Little Prince', Antoine de Saint-Exupéry reveals the mind of a good engineer when he envisions tiny planets with one occupant each, for whom everybody else is of the same *type*. He thinks in the same foundational abstractions on which many complex modern frameworks are built.

On asteroid 325 there is only a king, and to him everybody else is a subject:

> "Ah! Here is a subject," exclaimed the king, when he saw the little prince coming

In Linux, or rather from the point of view of Linux, everything is a file: folders are files that list other files, programs are executable files, and hardware devices are files that you can read from and write to. In HTTP, everything is a resource addressed through standard verbs: GET, POST, DELETE… In Erlang, the programming language of choice for scalable concurrency, everything is a process (as you would be inside a Hoapa protocol).

And not just in engineering: in living organisms every constituent is a cell, in Legos everything is a brick. They all standardize the container to liberate the content. This *unityping* of the right abstraction lets a system tame its own constitutive complexity.

Geographer/explorer separation of concerns

The little prince visits several of those population:1 celestial objects, and one of them is very close in spirit to Hoapa protocols: asteroid 330, whose lone resident is a geographer. Hoapa and this geographer have the same task: to chart their world, and the same inputs: explorers. And both will produce "voluminous books", except the Hoapa ones will welcome a lot more diversity and nuance while being a lot more accessible and pedagogical than the old fashioned atlases.

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Hoapa protocols are autonomous computer programs that can emulate the geographer to produce a decentralized form of citizen science. And they are in the same empty/waiting state at the moment: (almost) ready to do their work, but without input to work with.

Obviously, with those allegories, Saint-Exupéry is criticizing modern tendencies for uniformity, non-humanity and non-experimentalism. Yet this is exactly what Hoapa embraces and uses to its advantage to achieve another kind of architectural harmony: separation of concerns. The protocol *reflects* reality but never *touches* it.

In order to enjoy all the unique pluralism, humanness and experiences that make life beautiful, any group of explorers should have a separate system in which to encapsulate the "dirty work" of geography, but also philosophy and all sciences, such that this enterprise and a life of freedom don't interfere with one another. Like the fox's rites in the story, separation of concerns gives each activity its proper time and place. (Fittingly, Hoapa's mathematical incarnation is a topos, a 'place' in ancient Greek).

Democratic evolutionary conversations for truth verification

The shape of this insulated system is not arbitrary. Once a designer has chosen the egalitarian and freedom-preserving path—unitype the participants, separate the map from the territory—the rest of the architecture lights up ahead like a landing strip at night. Across disciplines, it forms a very coherent and satisfying network of voluble and self-reflecting, conversation-inviting concepts. All in the name of fairness, and legitimacy.

Hoapa protocols therefore take in life-inputs and *process* them without turning either the input or its carrier into an object of *judgment*.

This is reflected in the contrasting ways the geographer of asteroid 330 and the community of explorers in Hoapa protocols verify truth:

- The geographer *looks inward and back* when presented with new information: first he decides what is worthy of interest, then orders inquiries into the life of contributors, and will later demand proofs of their claims. Truth is decided at the scale of the individual proposition, who comes under a scrutiny organized by the geographer himself.
- Hoapa *projects forward* and submits any new data to an informational ecosystem regulated by an evolutionary game, where randomly selected explorers are the referees. Truth is decided at the scale of the group, which uses its dynamics to filter out the mistaken and irrelevant as it refines and composes the remainder.

This evolutionary game is again unityped like Legos: to it every process is a *conversation*. Those are shaped exactly like the citizen assemblies/conventions of participative democracy: select a few citizens at random, give them ample time and resources to study their topic, let them write a document of their findings by consensus and, most importantly, take their work seriously and use it. (Randomness is here the most democratic strategy, since everyone is actually on the same footing.) This strategy is field-proven and has shown time and time again how men and women "in the street" perform extremely well when put in the right context. In a very real sense Hoapa has ethics baked-in at its base.

And this is preserved in the mathematics that composes those bricks of processing to scale gracefully to possibly infinite structures. In three major ways—evolvability, expressivity and multiplicity:

- Protocols are always evolving, not just in terms of what they produce but also as engineering constructions. The computer code they're made of is automatically and regularly subjected to modifications by conversations, organized just like the others but with the special mandate to edit the code if they see fit. (Protocols are managerless.)
- The documentation protocols produce also becomes the language they use. The concepts they map are given names with accessibility, pedagogy, interoperability and overall coherence as top priorities (following a "learn once, express everywhere" philosophy). Eventually every new member coming into the protocol is asked the same effort, minimized by design, to engage with its concepts.

- From their own perspective as engineering constructions, protocols are "aware of" their neighbors and "want to be part of" networks. They always "think of themselves" as members of a multiplicity. (The Yoneda philosophy, after arguably the most beautiful result in abstract mathematics which says mathematical objects remember their friends when they're separated).

In a nice duality with the unityping of participants, most of this composition is encapsulated in *type theories*. Mathematical structures which have also proven their worth in several fields, as a way to organize computation, meaning, existence, proofs or equality.

Interestingly, most of the mathematics we use have been discovered by an unusual geometer and obstinate ecologist who wrote volumes about the creativity of one's inner innocent child, Alexander Grothendieck. (They stand at the opposite end of the spectrum from the simplistic models used by mainstream economists to justify the exploitation of Nature and populations, or by some evolutionary biologists to deny the true complexity of ecosystems.)

Nutshell

In summary, Hoapa are based on a double separation of concerns:

- The *protocols* are separate from *reality*: they want to "tell its story" but in order to tell it faithfully they must never be part of it.
- The participants are all strictly *unityped* in the eyes of the protocol, no judgement at this stage but only the *welcoming* of complexity; while the voluminous documentation the participants produce collectively is in effect a multilayered *type theory*, the form it must take to adequately *render* the complexity processed by the system.

And beyond, those foundational principles unfold into a lush cascade of further separations, type theories and translations, emerging through the conversations and their orchestration by elegant mathematics, all in the service of optimizing the protocol's autonomy, decentralization, traceability, evolvability, expressivity, multiplicity, etc.

Convoluted

In 'The Little Prince', Antoine de Saint-Exupéry shows he thinks like a good engineer—with the same kind of foundational abstractions that Linux, HTTP or concurrent programming are based on—when he envisions tiny planets with one occupant each, for whom everybody else is of the same *type*.

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The shape of this insulated system is not arbitrary: the design of Hoapa is entirely specified by its foundational principles whose philosophical origin is at a nexus of theses such as the univocity of Being, presuppositionlessness or perspectivism (à la Duns Scot, Hegel and Deleuze respectively, where univocity ≅ unityping). All in the name of fairness, and legitimacy.

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0.1 Roadmap

The sense of my work is twofold:

- To make the system fully ready to receive its first explorers (the implementation)
- To make the system legitimate enough to motivate participation (the verifiable theory)

see note

The legitimacy comes from the minimalist -- separations of concern.

-> separations of concern: "external" - protocol (dirty work) | reality (freedom to explore) ; "internal" - unityped agents | typed documentation (only ideas get "put in boxes", not people)

, whose success has been credited to a "universal sense of ethics", which is because this work is about attachment
-> theory of attachment, how the protocol eventually produces mature nodes to act as secure bases for further exploration

used those absurd planets to

as such (no looking back), And they aren't restricted to geography, they're interested in whatever the explorers want to talk about. -> hoapa means everywhere.

They "move information around" without ever knowing what it is.

Or modularity, how to design subsystems that will each perform one task well so that they form _ systems when composed together. To make simplicity local and complexity collective.

It deals only with containers, without ever opening the boxes.

(first theorem is about the bridge between the explorers and the orchestration of conversations).

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WITH REFS

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everything is a resource that you GET, POST, DELETE and so on. In Erlang, the programming language of choice for scalable concurrency, everything is a process [Hewitt] (just like you will be, if you ever join a Hoapa protocol). In living organisms every constituent is a cell [phenotypes], in Legos everything is a brick [philo]. They all standardize the container to liberate the content. It's this *unityping* of the right abstraction that lets a system tame its own constitutive complexity.

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